AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2011

Physics Paper I

Time allowed: 35 minutes Marks 25

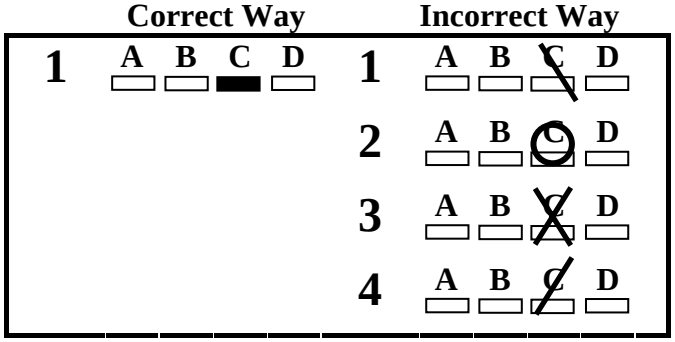
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.

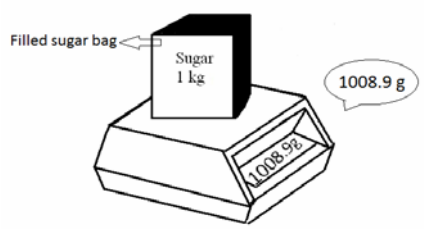


5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

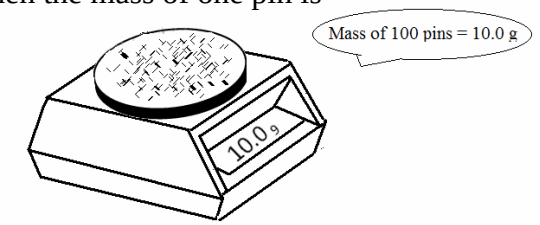
7. You may use a simple calculator if you wish.

1. The mass of the empty sugar bag in the given figure is



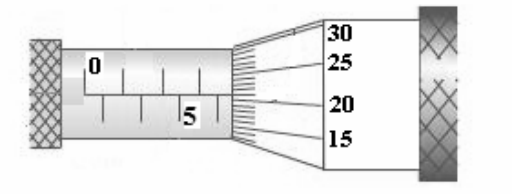
A. 8.9 g
B. 10 g
C. 100 g
D. 800 g

2. In the given figure, if all pins are of the same mass, then the mass of one pin is



A. 0.001 g
B. 0.01 g
C. 0.1 g
D. 1 g

3. If the least count of the given screw gauge is 0.001 cm, then the total reading will be



A. 0.00721 cm
B. 0.0721 cm
C. 0.721 cm
D. 7.21 cm

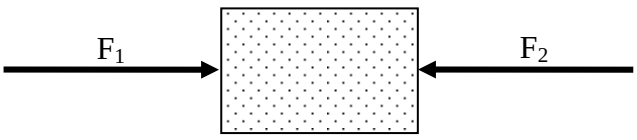
4. The opening and closing of a door is an example of

A. spin motion.
B. vibratory motion.
C. projectile motion.
D. translatory motion.

5. A free falling body is an example of

A. circular motion.
B. rotatory motion.
C. vibratory motion.
D. translatory motion.

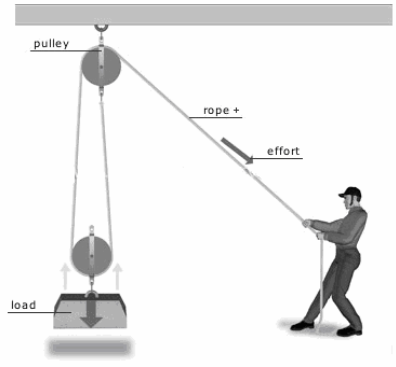
6. Two equal and opposite forces are acting on a body.



Which of the following quantities CAN NOT be zero in the above diagram?

A. Mass
B. Velocity
C. Momentum
D. Acceleration

7. A person supports a mass of 20 kg suspended by a rope. What is the resultant force acting on the mass?

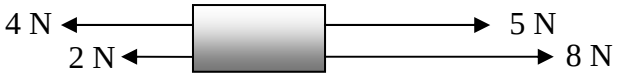


A. 0
B. 0.2 N
C. 2 N
D. 10 N

8. A force produces change in all of the following EXCEPT

A. mass.
B. shape.
C. speed.
D. length.

9. If four forces are acting on the body in the given diagram, then the magnitude of the resultant force will be

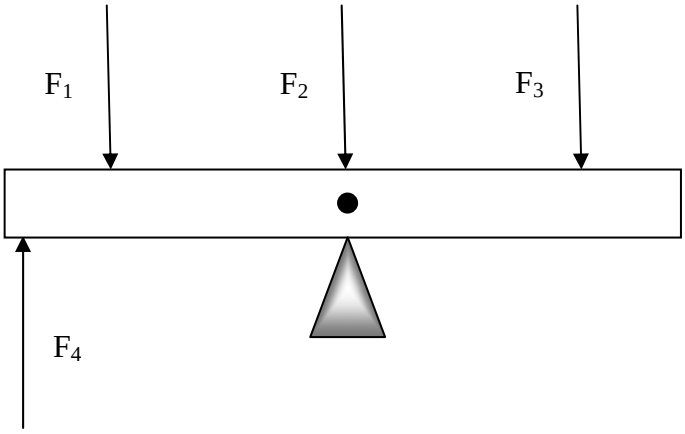


A. 1 N
B. 6 N
C. 7 N
D. 19 N

10. If a vector $\vec{A}$ has rectangular components $\vec{A}x = 4$ and $\vec{A}y = 4$, then the angle between vector $\vec{A}$ and *x axis* will be

A. 0º
B. 45º
C. 90º
D. 180º

11. Four different forces of the same magnitude are acting on the given meter scale. Which force is producing the maximum torque?



A. F_1
B. F_2
C. F_3
D. F_4

12. The necessary acceleration for a satellite orbiting the earth in provided by

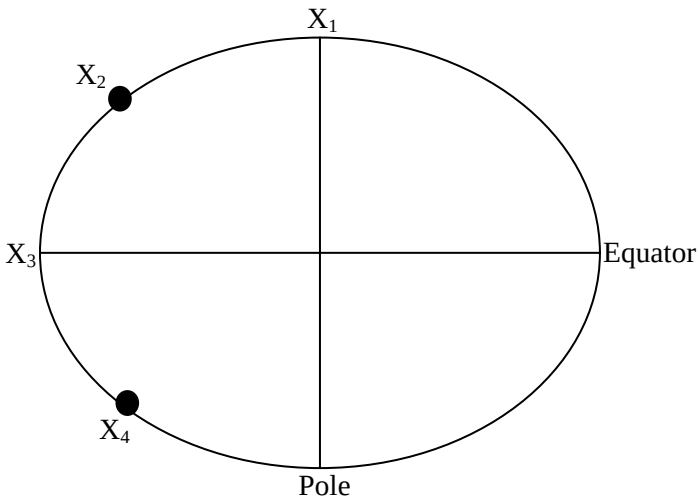
A. frictional force.
B. coulomb's force.
C. centrifugal force.
D. gravitational force.

13. If the mass of an object is 1 kg and the value of g is 10 m/sec^2, then the weight of the object will be

A. 1.0 N
B. 9.8 N
C. 10.0 N
D. 10.1 N

14. In the given diagram of Earth, the value of g will be least at

A. X_1
B. X_2
C. X_3
D. X_4



Earth

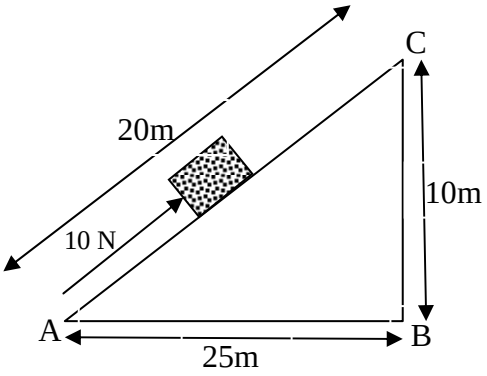
15. According to the data given in the table the work done by an object will be

Mass	Acceleration	Displacement
10 kg	2 m/s^2	3 m

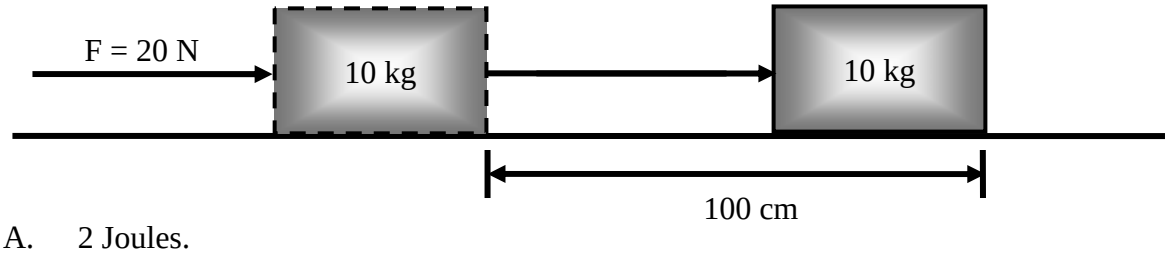
A. 5 J
B. 20 J
C. 40 J
D. 60 J

16. In the given diagram, if the block is moved from point *A* to point *C*, then the amount of work done will be

A. 100 J
B. 200 J
C. 250 J
D. 300 J

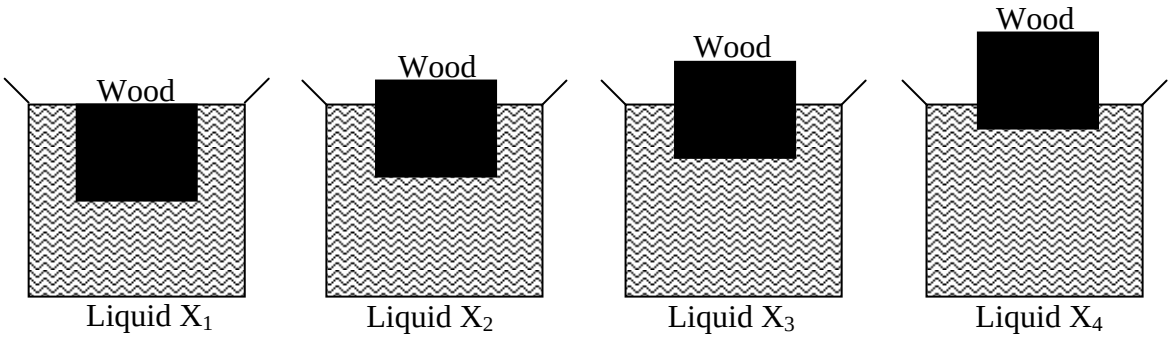


17. In the given diagram, the amount of work done by the object is



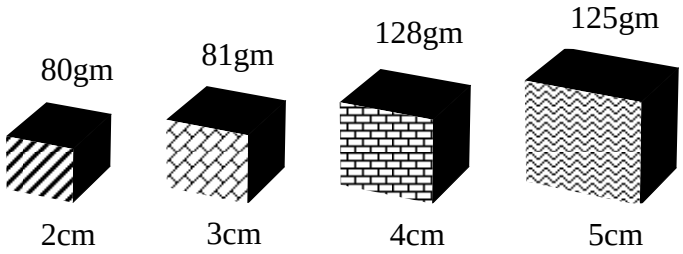
A. 2 Joules.
B. 20 Joules.
C. 200 Joules.
D. 2000 Joules.

18. A piece of wood is shown floating in four different liquids. Which liquid is denser than the others?



A. X_1
B. X_2
C. X_3
D. X_4

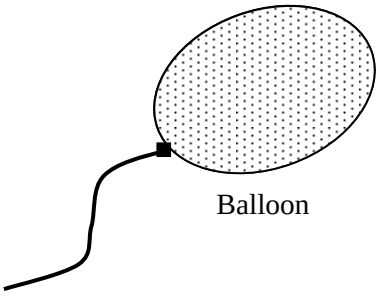
19. Which of the following cubes has the maximum density?



(A) (B) (C) (D)

20. If a sealed balloon, filled with a constant quantity of gas, is cooled down, then how will the listed properties of the gas behave?

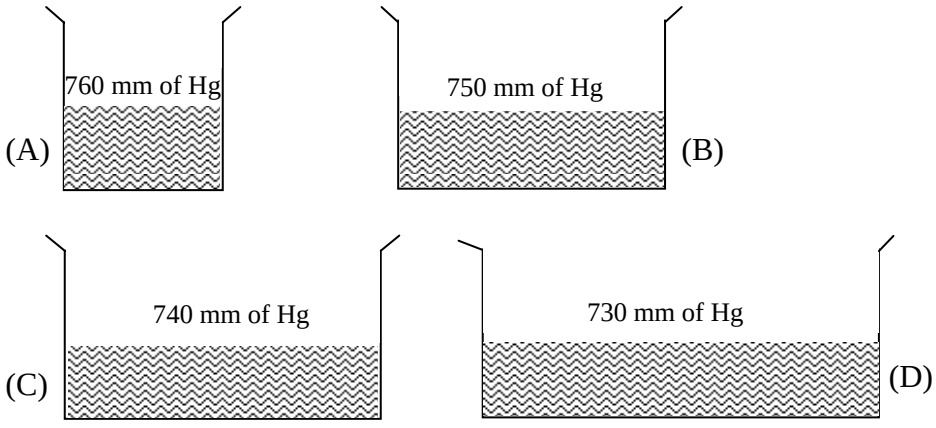
	Mass	Volume	Density
A	decreases	stays the same	increases
B	increases	stays the same	decreases
C	stays the same	decreases	increases
D	stays the same	increases	decreases



21. Absolute zero is the temperature at which the molecules of a body have

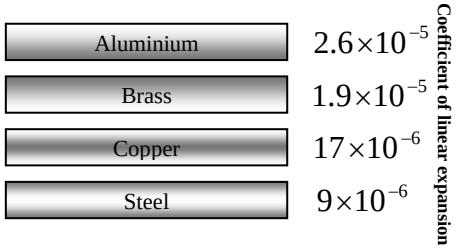
A. zero energy.
B. kinetic energy only.
C. potential energy only.
D. both kinetic energy and potential energy.

22. If the same quantity and same nature of liquids are present in the given vessels, then in which vessel will the liquid evaporate fastest?



23. If four metallic rods of the same length are heated equally, which will expand more?

A. Aluminium
B. Brass
C. Copper
D. Steel



24. A bimetallic thermostat is used to maintain temperature. On which of the following principles does it work?

A. Thermal radiation
B. Thermal expansion
C. Thermal convection
D. Thermal conductivity

25. Which of the following characteristics should be present in a surface to protect it from infrared radiation?

A. Poor absorbers and poor emitters
B. Poor absorbers and good emitters
C. Good absorbers and poor emitters
D. Good absorbers and good emitters

END OF PAPER

Please use this page for rough work